



ROSE: A Universal Neural Grammar

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ABSTRACT

Processing natural language syntax requires a negotiation between symbolic and subsymbolic representations. Building on the recent representation, operation, structure, encoding (ROSE) neurocomputational architecture for syntax that scales from single units to inter-areal dynamics, I discuss the prospects of reconciling the neural code for hierarchical syntax with predictive processes. Here, the higher levels of ROSE provide instructions for symbolic phrase structure representations (S/E), while the lower levels provide probabilistic aspects of linguistic processing (R/O), with different types of cross-frequency coupling being hypothesized to interface these domains. I argue that ROSE provides a possible infrastructure for flexibly implementing distinct types of minimalist grammar parsers for the real-time processing of language. This perspective helps furnish a more restrictive 'core language network' in the brain than contemporary approaches that isolate general sentence composition. I define the language network as being critically involved in executing specific parsing operations (i.e. establishing phrasal categories, tree-structure depth, resolving dependencies, and retrieving proprietary lexical representations), capturing these network-defining operations jointly with probabilistic aspects of parsing. ROSE offers a 'mesoscopic protectorate' for natural language; an intermediate level of emergent organizational complexity that demands multi-scale modeling. By drawing principled relations across computational, algorithmic and implementational Marrian levels, ROSE offers new constraints on what a unified neurocomputational settlement for natural language syntax might look like, providing a tentative scaffold for a 'Universal Neural Grammar' – a species-specific format for neurally organizing the construction of compositional syntactic structures, which matures in accordance with a genetically determined biological matrix.

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1. Introduction

Humans possess an extraordinary ability to assemble individual words into larger structures. This *syntactic* ability has been claimed to be intimately related to the origins of our species (Everaert et al., 2015; Hagoort, 2023), but our lack of understanding concerning how syntactic information is neurally encoded remains an obstacle for clinical and neurosurgical treatment (Riva et al., 2022). Although we are currently living in perhaps the most experimentally diverse time in cognitive neuroscience, there also comes a need for theoretical reassessment (Adger, 2025; Murphy, 2025; Piantadosi, forthcoming; Senkowski & Engel, 2024). Neurolinguists increasingly run the risk of deploying sophisticated new tools *sans* hypotheses and being labeled the 'stamp collectors' of cognitive neuroscience (Bernal, 1939, p. 9).

A major topic of current research concerns whether we can build more principled relations between natural language syntax and neural mechanisms driving statistical inferences. Most accounts either place heavy

emphasis on symbolic knowledge (e.g., Chomsky, 2013, 2023; Chomsky et al., 2019, 2023; Friederici, 2017; Murphy, 2023, 2024; Murphy, Holmes et al., 2024) or predictive processing (e.g., Caucheteux et al., 2023; Kwiatkowski et al., 2012; Schrimpf et al., 2021; Zhou et al., 2025), without a means for integration (but see Martin, 2020; Sartran et al., 2022; Yang et al., 2017; Yeaton, 2025). In this article, I will assess an emerging model of the neural infrastructure of syntax (ROSE; Murphy, 2024) and assess how statistical properties of language can be integrated with compositional processing. This will be achieved by reference to a frontotemporal *symbolic low-frequency phase code* interacting via cross-frequency coupling (Canolty & Knight, 2010; Canolty et al., 2006, 2010; Weissbart & Martin, 2024) with a series of local *probabilistic inferences over lexico-semantic content*, with the latter being implemented via spike-phase coupling assembling bundles of linguistic features and which can emerge into 'dynamical motifs' (Driscoll et al., 2024). The low-frequency phase

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code actively gates and modulates local spiking assemblies that represent lexical items (see also Pescatore et al., 2025). My proposal mirrors trends in neurosymbolic AI where both discrete operations and continuous dynamics are combined (Dhanraj & Eliasmith, 2025), and trends in cognitive neuroscience highlighting how the brain's activity time courses reflect underlying network-level computational mechanisms (Oby et al., 2025).

First, I will review the ROSE model (Section: 'Prospects for ROSE') then discuss how probabilistic processing can be implemented within it (Section: 'Structure Meets Statistics'). My goal here is to take a model that is explicitly devised to handle recursive, hierarchical symbolic phrase structures (ROSE) and develop an interface with broader components of language.

2. Prospects for ROSE

In this section, I will introduce the essential components of ROSE before reviewing new directions for expansion and refinement based on recent experimental research.

The ROSE (Representation, Operation, Structure, Encoding) neurocomputational architecture for syntax provides an explicitly *symbolic* account of how humans generate hierarchical phrase structures (Murphy, 2024). The genesis of this model can be found in Murphy (2015b) and subsequent publications (e.g., Murphy, 2016, 2020b) (qualifying the claim in Tavano et al. (2023) that the work of Meyer et al. (2020) 'de facto prefigur[ed] an oscillatory theory of language' in Murphy (2020b)). ROSE is built on the premise that a core (and deeply puzzling) component of language is its compliance with *structure-dependence* over linear rules: '[W]e ignore the simple computation on linear order of words [adjacency], and reflexively carry out a computation on abstract structure' (Chomsky, 2023). This principle is available very early to infants (Perkins & Lidz, 2021; Shi et al., 2020). ROSE is in line with the mathematically proven feasibility of unobservable entities, like mental representations, being appropriate objects for scientific study (Piantadosi & Gallistel, 2024).

ROSE is based on the assumption that the basic data structures of syntax are atomic features (Collins, 2024); types of linearly readable mental representations (R) that are coded at the single-unit and ensemble level. Operations (O) transforming these units into manipulable objects accessible to subsequent structure-building levels are coded via high-frequency broadband γ activity. Recursive structural inferences (S) over hierarchical phrase structure (providing instructions for things like headedness in linguistic sets (determining phrase category) and tree-structure node counts) are derived by forms of low-frequency phase synchronization and

phase-amplitude coupling (PAC) (e.g., low-frequency phase synchronization varies systematically with linguistic structure composition; Brennan & Martin, 2020). ROSE provides forms of causal mechanisms that can coordinate each of these levels: Spike-phase/LFP coupling ensures the operations at O coordinate spike firing of atomic representations at R. Various types of phase-amplitude coupling over left frontotemporal language sites provide a top-down code for reading out the cluster of representations maintained by spike-phase coupling, connecting O to S (see also Mark et al., 2024 for evidence that cross-frequency coupling provides a biomarker for early stroke recovery).

Depending on the lexical item in question, different spikes from relevant cortical regions will be coordinated by spike-phase coupling (e.g., posterior middle temporal cortex for abstract word features; inferior parietal cortex for eventive features; anterior temporal lobe for object features; ventrotemporal cortex for face and place features; inferior frontal cortex for more formal and function-word related features). Some of these features relevant to syntax may generally localize to distinct sites (Morgan et al., 2025). To establish headedness, multiple different θ cycles representing feature-bundles (generating a morpheme) have their amplitude coupled to the phase of δ , with the θ rhythm that exhibits the highest strength in phase-amplitude coupling determining which morpheme is the head. Effectively, multiple θ bursts nested in the same δ cycle 'compete': the burst with the highest phase-amplitude coupling strength is temporally centered on the δ trough. A newly-built δ - θ complex can itself be merged with prior complexes via δ phase-phase coupling. The new δ cycle couples its phase to the preceding δ trough (phase-reset), yielding a serial 'push-reset-push' dynamic. Every time δ phase resets, the current θ - γ coded object is pushed into memory via a short β burst, and a new δ complex starts to encode the higher-level structure. Because each complex is identical in circuitry, the system can, in principle, recurse indefinitely – bounded only by workspace limits reflected in α -band interference when too many δ complexes are simultaneously active. For a mathematical formalization of some of these aspects of ROSE pertaining to phase synchronization, linking directly to algebraic models of syntax, see Marcolli and Berwick (2025).

Data-driven CA1 models show that two spatially separated θ - γ generators can coexist and compete for phase space with the same δ pacemaker (Ponzi et al., 2023); which ensemble dominates is set by the strength and directionality of their coupling with δ . The θ - γ generator that enjoys the most privileged excitability window with δ drives compositional updates downstream. If the head-ensemble has the largest δ -coupling, its phase-

gradient could set the angle of the traveling wave that carries semantic features into distant cortex, mirroring how a syntactic head projects its features through the tree. Frontotemporal traveling waves transfer the hierarchical phrase structure representations at S to active workspaces (the E level) in relevant cortical memory sites, the computational scratchpad for natural language. ROSE enables potentially unbounded recursion until rising fronto-parietal α signals that the syntactic workspace (δ - θ dynamics) is full, while frontotemporal traveling δ waves ferry each completed complex to working memory buffers. Essentially, temporal gating in this delta-theta-gamma complex encodes non-associativity (see Figure 2 and discussion below). Since each MERGE happens in a different δ -phase segment, the history of composition is ordered in time: $((W_1 W_2) W_3) \neq (W_1 (W_2 W_3))$. We might assume that each δ -cycle ends with a β -mediated 'commit' burst (Lundqvist et al., 2024) that silences the γ carriers for the daughters, ensuring the composite enters the next workspace step as an indivisible unit. Because headedness is selected by mutual information PAC strength rather than presentation order, the system is commutative, yet the β commit burst freezes the set, preserving non-associativity. As such, blocking the β -commit burst should lead to illicit syntactic re-groupings (associative errors) detectable as spurious γ re-activations of daughter morphemes. The issue of commutativity can be approached here also by assuming that the early stages of ROSE only need to bundle features without strictly determining order via their read-out at higher levels (Figure 1). One issue for future research concerns how to *guarantee* that this neural code breaks associativity, rather than simply breaking it 'just in case' the representation is a linguistic one; see Marcolli and Berwick (2025).

Low-frequency rhythmic neural activity has been found not just to gate and regulate but also communicate specific stimuli content (Stecher et al. 2025), and under ROSE it communicates structural linguistic content. These elements are represented in Figure 1. Given that under many current accounts 'architectures for processing symbols seem decidedly unbiological' (Piantadosi, 2021), ROSE provides a path toward biological plausibility. Some have argued that phase-amplitude coupling is only suited to encoding sequential relations (Coopmans et al., 2023). Yet, as argued in Kazanina and Tavano (2023b) and Murphy (2020b, 2024), phase-amplitude coupling is far from being limited to this.

ROSE can model *vertical* aspects of syntax, like the recursive hierarchical formation of phrase structures. This involves explicit symbol manipulation, of the traditional kind (Kolers & Smythe, 1984; Marcus, 2001; Murphy & Leivada, 2022). We can also, in principle,

accommodate how language-specific conditions on MERGE – like 'theta theory', valency and properties of lexical selection – have a *directive* effect on MERGE (e.g., verb transitivity) (Chomsky et al., 2023), since lexico-thematic constraints are realized as feature complexes at the R level, synchronized and bundled at the O level, thereby regulating which phrase structures the S level can build. The O level would filter and organize lexical constraints emerging from R. Yet, further refinement is needed to incorporate relational and *horizontal* types of information (e.g., order/adjacency, filler-gap relations, linear encoding of morphosyntactic agreement). Here, I demonstrate the feasibility of integrating associationist and predictive neural mechanisms into ROSE.

My approach aligns with some previous proposals (Hagoort, 2023; Kanerva, 2003; Martin & Doumas, 2020; Plate, 2003; Smolensky et al., 2022) and is similar to Marcus's (2001) approach to integrating symbolic and connectionist models of cognition. Marcus argued for a hybrid, neurosymbolic architecture that incorporates meritorious aspects of both symbolic (rule-based) and neural network (connectionist) models. This might involve incorporating symbolic systems (e.g., knowledge graphs, logic programming) with neural networks to enhance reasoning; many aspects of linguistic semantics, like the parsing of negation (Coopmans et al., 2024), seem to require forms of knowledge that fall outside of connectionist and embodied accounts.

2.1. A need for symbols

Forms of symbolic and subsymbolic linguistic information seem to be represented across a broad frontotemporal language network (Lopopolo et al., 2021; Murphy, Rollo, et al., 2024). Posterior temporal cortex is often seen here as the critical node in the core syntax network (Biondo et al., 2024; Fahey et al., 2024; Matchin & Hickok, 2020; Matchin et al., 2024; Murphy et al., 2023; Yu et al., 2024), and may be the earliest region to jointly resolve vertical and horizontal dimensions in auditory phrase parsing (Murphy, Rollo, et al., 2024). An integrated measure of processing costs pertaining jointly to these dimensions can explain temporal lobe activity in fMRI (Li & Hale, 2019).

Along with many others, I have recently argued that the success LLMs have displayed for modeling lexico-semantic information and dependency relations is likely a factor of their competence for linearized syntactic information (Leivada, Dentella, et al., 2023; Leivada, Murphy, et al., 2023; Linzen & Baroni, 2021; Murphy et al., 2025; but see; Marcolli et al., 2025 for a different perspective). I also note here that LLMs cannot, in principle, provide a 'theory' of linguistic competence (Alers-Valentín & Fong, 2024). It seems possible that LLMs do

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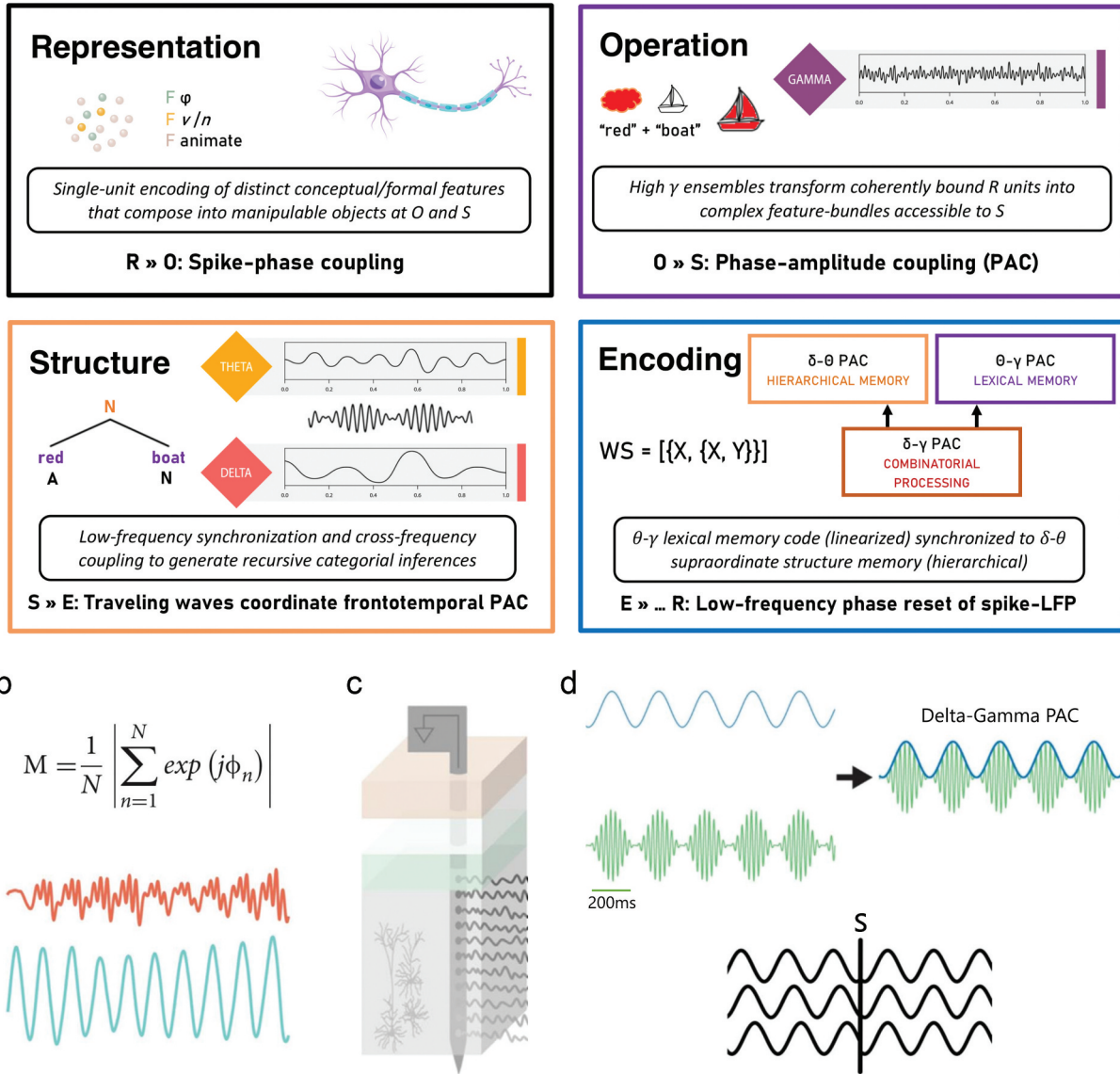


Figure 1. A representation of the ROSE neurocomputational architecture. (A) Boxes contain graphical depictions of the relevant organizational scale (top), a description of how each scale relates to the relevant component, and the putative mechanistic link between components (bottom text). (B) Spike-LFP coupling can be quantified by utilizing a phase-locking value (PLV) measure. PLV calculates the power of dependability or linking of LFP phases in spike times, by computing the angular summation between spikes to some low-frequency range LFP fluctuations. Here, N shows the number of spikes, ϕ_n is related to the instantaneous phase at the time of n th spike which is determined by Hilbert transform, and $\exp(j\phi_n)$ is the complex exponential function of ϕ_n . The amplitude of vector M ($|M|$) indicates the spike-phase coupling power and its angle ($\angle M$) shows the phases of LFPs in the time of spike occurrence. (C) A cartoon depiction of an intracranial probe detecting spike activity (horizontal colors from top to bottom: skin, bone, cerebrospinal fluid, grey matter with neuron representations; adapted from Sanchez-Todo et al. (2023), Fig. A1). (D) Top: a toy example of phase-amplitude coupling (PAC). Below: a stimulus (S) results in a phase alignment of ongoing neural oscillations.

indeed incorporate some version of vertical syntax (Diego-Simón et al., 2024) – potentially via their attention mechanism – but in a wholly different format from how the human brain represents vertical structure (Dentella et al., 2024). Hence, my position will be that artificial models should most reliably be used for exploring horizontal dimensions (see Supplementary Material).

Current evidence indicates that modern language models struggle with causal reasoning and directed exploration (Binz & Schulz, 2023), compositionality (Dentella et al., 2024; Lake & Murphy, 2023; Murphy et al., 2025), analogy and human-like abstractions (Mitchell, 2021), amongst other things (for related critical discussion, see Baggio & Murphy, 2024).

When deep language algorithms are shown to predict semantic comprehension from neural activity, they do so within parietal and medial temporal lobe structures (Caucheteux et al., 2022). In the case of Caucheteux et al. (2022), this alignment seems to be generated to a large extent by the long-distance dependencies within deep layers of GPT-2 (Radford et al., 2019), a system that uses unidirectional, forward attention, highly suited to horizontal linguistic information. When measures of surprisal were contrasted with measures of syntactic node counts during sentence comprehension, intracranial electrode sites sensitive to lexical surprisal were found to be generally distinct from electrode sites sensitive to syntactic structure (Nelson et al., 2017; see also Woolnough et al., 2023). Next-word prediction has been shown to be insufficient on its own to account for human syntactic processing (Huang et al., 2024; see also Fodor & Ferreira, 1998). Even the predictive coding framework is currently being reassessed in terms of how it may be ‘more cognitive’ than previously assumed (Gabhart et al., 2025). I provide in Supplementary Material a brief critique of the current state of the art in deep language models, as a form of converging evidence for *why connectionist models alone* are insufficient to capture the neural code for syntax.

What, then, is the alternative? Symbols famously exhibit a number of advantages over continuous representations (Marcus, 2001), such as robustness to noise, reduced bandpass, flexibility, distinguishability, unbounded constructability, compactness, and simplification. The following section will provide key details concerning this symbolic knowledge of language.

2.2. Foundations in syntax

The symbolic architecture of ROSE takes as its target the operation MERGE and its associated algorithm for ‘labeling’ (assigning categories to) hierarchically formed magmas of syntactic objects (Chomsky, 2013; Hornstein, 2009; Marcolli et al., 2025; Murphy, 2025; Pan et al., 2024). This algorithm has been claimed to be orchestrated by concerns of computational efficiency, minimal search, and other constraints naturally suited to symbolic systems (Murphy, 2015a, 2015c, 2025; but see also Goto & Ishii, 2024). This constrains the otherwise unbounded nature of MERGE (Ginsburg, 2024) and establishes ‘headedness’ (see Figure 3), another clear example of symbolic knowledge. Labeling is a means by which linguists can explain, from first principles, grammaticality of structures, the distribution of phrases (e.g., noun phrases), and even aspects of language change over centuries (van Gelderen, 2023). The types

of symbolic principles pertaining to MERGE are claimed in Murphy (2024) to be necessary for building a cognitively plausible neurobiology of syntax (see also Uddén et al., 2020). The mechanisms of ROSE are claimed to be strong candidates for implementing processes that afford explicit syntactic representations, given the well-documented ‘managerial’ and ‘orchestrating’ role for oscillatory dynamics (Kazanina & Tavano, 2023a; Mendoza-Halliday et al., 2024).

MERGE can be formalized as follows (WS = workspace; P/Q = workspace objects; X = additional elements):

$$WS = \{P, Q, \dots\}$$

$$\text{MERGE}(P, Q, WS) = WS' = \{\{P, Q\}, X_1, \dots, X_n\}$$

This operation takes two objects inside a workspace and maps them to a new workspace. However, this formulation has been challenged recently, since it relies on naïve set theory, which comes with virtually no constraints. MERGE has recently been formulated as a free non-associative commutative magma (Marcolli et al., 2025) that generates nonplanar trees (and provides much-needed structure and constraints, in contrast to older models using naïve set theory). A planar embedding only arises during externalization via sensorimotor compression and linearization (Kayne, 1994). In this more recent formulation (Marcolli et al., 2025), the set $\mathcal{SO}$ of syntactic objects is the free, non-associative ($_{na}$), commutative ($_{c}$) magma over the set $\mathcal{SO}_0$:

$$\mathcal{SO} = \text{Magma}_{na,c}(\mathcal{SO}_0)$$

With the binary MERGE operation:

$$\mathfrak{M}(\alpha, \beta) = \{\alpha, \beta\}$$

Figure 2 provides an outline of the MERGE-based syntax I assume for ROSE.

Both Murphy (2015a) and Hornstein (2024) defend the ‘Labeling Hypothesis’ that labeling constitutes the evolutionary novelty of language, since labeling is ultimately the operation critical to closing MERGE and therefore delivers a recursive system of unbounded hierarchy. Labeling provides rigid categorization (dominance by type) and the ability to saturate dyadic concepts, going beyond monadic concepts (Pietroski, 2018), giving us different kinds of ‘things’ to think about. It is also labeling that permits compositional semantics, since it produces a representational result that cannot be reduced to its parts (Hornstein & Pietroski, 2009). It allows us to form *inductive definitions*, which for Hornstein (2024) is the major contribution of labeling. Relatedly, Carey (2023) argues that ‘there is no good evidence for nonlinguistic deductive reasoning involving the disjunctive syllogism,’ and that ‘[a]nimals and pre-linguistic children probably do not make logical inferences.’

I have been motivated here by the more general proposal ‘to use the computational demand of linguistic structure building to constrain possible neural mechanisms’

A Minimal Syntax for Natural Language

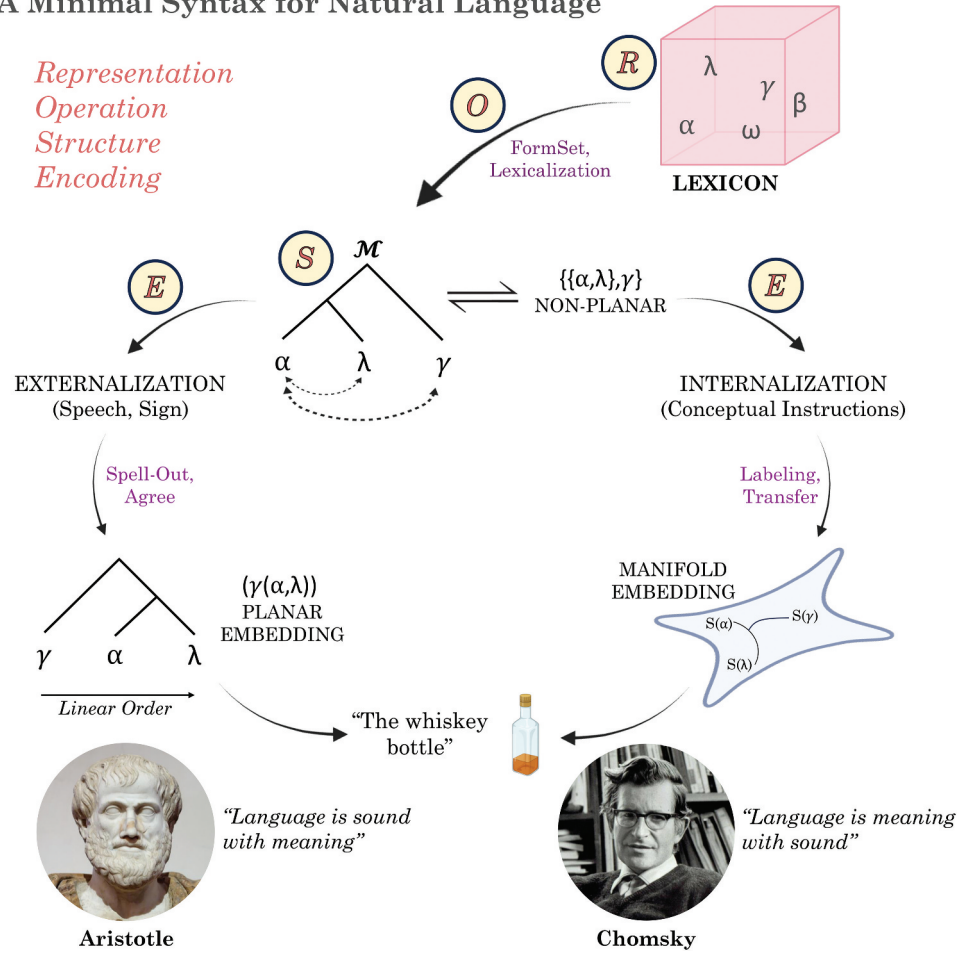


Figure 2. A representation of MERGE-based syntax. This model is strictly atemporal (i.e., a computational-level model), but associated levels of ROSE are linked to different stages of derivations, which will be further embedded in real-time parsing algorithms below (see Collins & Stabler, 2016; Marcolli et al., 2025). During internalization, structures are mapped to distinct conceptual systems (likely ‘core knowledge systems,’ Spelke & Kinzler, 2007) with their own distinct manifold embeddings to generate the concept *whiskey bottle*. Lexical elements (α, λ, γ) are mapped to a workspace via MERGE ($\mathcal{M}$), which is then sent to interpretive systems and categorized (‘labeled’), and/or are ‘spelled-out’ at externalization systems for speech, gesture, etc. For example, ‘the whiskey bottle’ would be formed via $\{\text{whiskey}, \text{bottle}\}$ being merged with $\{\text{the}\}$, to form a hierarchical object that is externalized in different orders depending on the language in question (i.e., not all languages adhere to the English ‘determiner noun’ order, but all human speakers converge on the same coordinates in semantic space for the phrase). Basic set-formation (FormSet), transfer to the interfaces (Transfer, Spell-Out), and Agree vs. Labeling are outlined for the interested reader in Chomsky et al. (2023). This formulation of MERGE provides potential (though not strict) constraints on what the neural code for syntax should look like. The quotations at the bottom contrast views on the design of language that centralize properties of externalization versus conceptualization (see Chomsky, 2011 for discussion). Portrait photographs reproduced from creative commons (Wikimedia Commons; commons.wikimedia.org): public domain photo of marble bust of Aristotle (left). Public domain photo of Chomsky from Andrea Womack (c. 1973) (right).

(Ding, 2023). Yet, I have discussed intrinsically atemporal formal accounts. The following sections will move toward concerns of algorithmic implementation, which will allow a more explicit connection with probabilistic processing.

2.3. Defining the language network

So far, I have defined some core syntactic computations. How can we use these proposals to constrain our definition of the brain’s language network?

Labeling and MERGE deliver hierarchical syntactic knowledge, while other operations like Agree (checking morphosyntactic ‘agreement’ between grammatical features) are thought to emerge due to non-linguistic sensorimotor ‘interface’ constraints (Murphy & Shim 2020; Narita, 2014). It is within this latter domain that statistical learning has been shown to be helpful in resolving agreement relations (Diego-Simón et al., 2024). Yet, cognitive neuroscience research typically only offers rather general characteristics for syntactic composition.

In response to this, I provide a more precise definition of the brain's 'language network' (Figure 3). This defines the language network as being responsible for establishing the core pieces of syntactic information: (1) Headedness, (2) Node structure, (3) Dependencies, and (4) Lexico-semantic features. This interfaces with systems responsible for the statistical properties of morphemes. When networks responsible for symbolic features interface with external networks (multiple demand, default-mode, etc.) this can be thought of as forming the *extended language network*.

The core language network is the set of cortical and subcortical regions and their interactions that selectively compute and represent symbolic operations proposed by minimalist grammar parsers: Establishing headedness (Wang et al., 2025; Zhao et al., 2025); Performing incremental node-counting operations (Desbordes et al.,

2023; Woolnough et al., 2023); Encoding and retrieving lexico-semantic and morpho-syntactic features; Resolving grammatical dependencies (agreement, movement, binding) (Hale et al., 2022).

This more restrictive definition of the language network mitigates the major issues of 'sentences > word-lists' activation maps and Jabberwocky-style localizers, including their inability to account for working memory, cognitive control, attention, world knowledge and situation model construction effects (Murphy & Woolnough, 2024). Indeed, it would be odd to propose a single experiment as the 'attention localizer', or the 'working memory localizer' – no single experimental paradigm can sufficiently isolate language-specific representations and computations. Mapping the neural signatures of isolated syntactic parsing features alongside statistical features can readily be achieved through

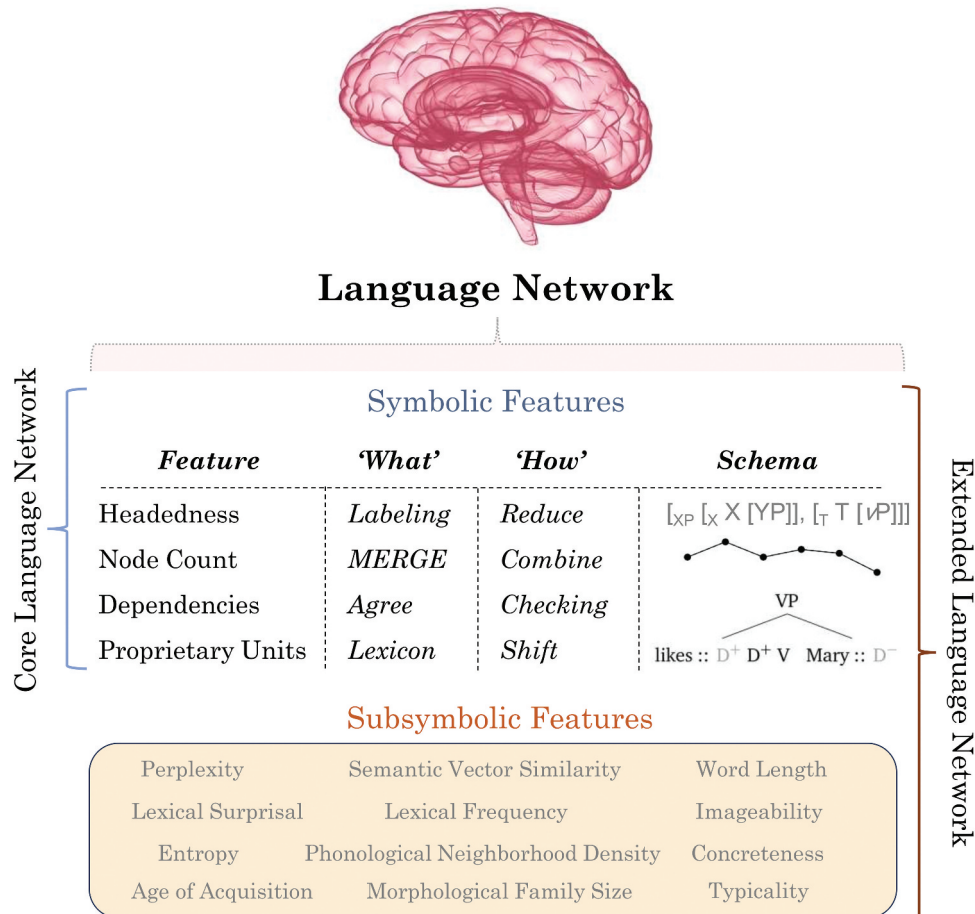


Figure 3. A definition of the human language network. Core language network characterized by selective involvement in MERGE-based parsing operations and how these align with computational-level accounts (Figure 2). For the tree structure representing 'likes Mary,' checked features are grayed out. With respect to the range of possible subsymbolic, continuous and surface-oriented statistical features, some of these would be derived from, e.g., GloVe vectors and English Lexicon Project-derived statistics (for representative examples of approaches to modeling subsymbolic feature spaces, see Forseth et al. 2021; Khanna et al. 2024; Woolnough et al. 2021, 2024; Woolnough & Tandon 2025). For an interesting alternative approach that defines the language network in less computationally explicit terms, potentially capturing dynamics overlooked here, see Fedorenko et al. (2024; 2025).

a combination of constrained experimental and naturalistic language processing approaches, which allow for each of these distinct parameters to be factored into relevant analyses. In comparison, recent research that vectorizes symbolic features often includes parts of speech, affixes and syntactic dependencies (Goldstein et al., 2025), but these are not tied to any specific model of language, nor any parsing model. Needless to say, the framework I offer here is in direct opposition to the hypothesis that syntactic parsing is essentially derived from semantic and statistical cues (Frank & Christiansen, 2018). It is likely that assuming a less computationally specific framework for the language network explains why some other analyses have led to claims about broad portions of the superior temporal sulcus (STS) being in the core language network (Shain & Fedorenko, 2025), rather than critical sub-portions of it that more specifically code for syntax-semantics.

As an initial cartographic guide beyond this formal definition, Figure 4 provides a general network depiction for the syntax network, which does not operate in isolation but interacts closely with working memory networks (e.g., dorsolateral prefrontal cortex) and cognitive control networks (e.g., frontoparietal attention systems). This network, though not exhaustive, characterizes the roles of particular regions based on recent

neuroimaging, intracranial and lesion studies (Baggio 2018; Beber et al. 2025; Chalas et al. 2024; Chang et al. 2015; Coopmans and Zaccarella 2023; Desbordes et al. 2023; Friederici 2017; Gao et al. 2025; Kazanina and Tavano 2023a; Jeong et al. 2025; Li et al. 2024; Maran et al. 2022; Matchin et al. 2022; Matchin and Hickok 2020; Morgan et al. 2025; Murphy 2020b; Murphy, Rollo, et al. 2024; Otani et al. 2025; Schoffelen et al. 2017; Woolnough et al. 2023).

This perspective leads to the hypothesis that the language network will exhibit a specific and characteristic cytoarchitectonic, histochemical, and connectivity architecture, which determines its representational make-up.

2.4. ROSE as a plausible model for syntax

I have so far defined core properties of linguistic computation and used this to generate a specific definition of the language network. I will now review newly-emerging empirical support for ROSE, which includes a number of suggestive directions for how to think about symbolic-connectionist modeling of language.

Consider first the study reported in Weissbart and Martin (2024). These authors examine phase-amplitude coupling in MEG during speech perception and

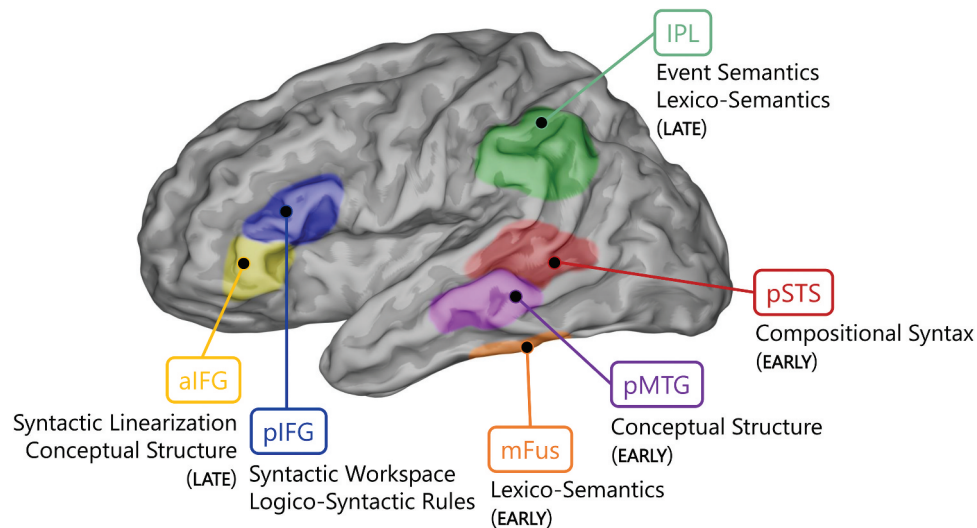


Figure 4. A basic representation of the brain's (broad) syntax network. Schema plotted on a semi-inflated N27 cortical surface (Pieters et al., 2013), labeled according to whether effects are typically seen early (approximately 100-300ms) or later after relevant stimuli onset (e.g., phrase-marking word). This is not intended to be a mono-functional, monolithic representation, but rather an indication of which nodes in the extended language network are generally implicated in certain computations (e.g., pMTG is *not only* implicated in compositional syntax; IFG is *also* involved in parsing Information Structure). For example, joint IFG-PT (posterior temporal) θ -burst stimulation using TMS destabilizes hierarchical syntactic processing to a greater extent than IFG-SPL (superior parietal lobule) stimulation (Gao et al., 2025), and frontotemporal regions jointly contribute to sentence-level semantic composition (Woolnough et al., 2023). Note that this representation is not meant to be exhaustive, and excludes white matter tracts (Sánchez et al., 2023). Regional abbreviations: aIFG (anterior inferior frontal gyrus); pIFG (posterior inferior frontal gyrus); mFus (mid-fusiform cortex); pMTG (posterior middle temporal gyrus); pSTS (posterior superior temporal sulcus); IPL (inferior parietal lobe). For a broader perspective here on functional localization, see (Makin and Krakauer 2023).

‘challenge a strict separation of linguistic structure and statistics in the brain, with both aiding neural signal reconstruction.’ They argue that an integration process between statistics and structure might be achieved via cross-frequency coupling. They demonstrate a fronto-temporal wave propagating from posterior to anterior MEG sensors in δ and in the opposite direction for θ , which under ROSE encodes structural information in a workspace. $\delta - \beta$ and $\theta - \gamma$ phase-amplitude coupling in posterior temporal regions and the temporo-parietal junction was connected to syntactic node closures, while $\delta - \beta$ phase-amplitude coupling was linked to syntactic complexity (node depth) in these regions and in inferior frontal cortex, supporting parts of the putative neural code and localization assumed in Murphy (2024). Syntactic and statistical features were jointly represented in inferior frontal and anterior temporal cortices. This work by Weissbart and Martin (2024) represents an innovative and imaginative approach to capturing important dynamics of real-time language processing that do not sideline key compositional mechanisms.

Phrase-rate neural tracking, measured via evoked power and inter-trial phase coherence, has been found to be sensitive to the *type* of syntactic structure being parsed in speech (the position of the ‘head’ in two-word phrases) in scalp EEG (Zhao et al., 2024). This points to the possibility that the integration of symbolic and probabilistic knowledge interfaces at the level of oscillatory phase codes. In related work, Zhao et al. (2025) focused on the neural representation of headedness (labeling decisions), and demonstrated sensitivity via inter-trial phase coherence in two components, straddling the δ/θ (~1–6 Hz) and α/β bands (~10–17 Hz) in scalp EEG to the representation of the phrase head. This is achieved by a reactivation specifically of the head and not other phrasal elements during the parsing of subsequently connected adverbs to the verb.

In related work, other researchers are arriving at the conclusion that compositional phrase processing recruits a widespread functionally connected network, demanding a focus on potential phase codes rather than purely localizationist region-function associations (Kitazawa et al., 2025). Ten Oever et al. (2024) show that a phase code of the kind evident in processes like attention also orchestrates lexical-phonological processing in language, such that highly probable events are coded at lower excitability phases. Ambiguous words presented at different phases (via neural entrainment or extracting the phase from MEG) are interpreted as one or another word depending on the phase of presentation. These results support core claims underlying ROSE, such as the notion of a memory phase code for lexicality (Murphy, 2020b, pp. 172–173; see also Cross et al., 2025).

Dekydtspotter et al. (2024) directly test ROSE through examining multi-cycle *wh*-dependencies in L1 and L2 French speakers, and provide support via scalp EEG specifically for the O component. They then innovate on this component and show how O can also accommodate properties of relational grammatical processing. Other scalp EEG research points to different frequency bands being implicated in lexical and syntactic processing (Cometa et al., 2024).

Single cells in the hippocampus have recently been found to be sensitive both to individuals and pronouns referring to those individuals (Dijksterhuis et al., 2024), supporting the formulation of R (Murphy, 2024) under which clusters of units encode relevant sets of conceptual features that conspire to form the basis of lexical items. Single neurons in prefrontal cortex respond selectively to specific word meanings and reliably distinguish words from nonwords (Jamali et al., 2024), being sensitive also to context of use (see also Franch et al., 2025). Other research has exposed neurons within frontotemporal language cortices that respond in an amodal manner to basic linguistic input (Lakretz et al., 2024). In this study, middle temporal cortex showed a preference for questions over declarative sentences, potentially indicating some sensitivity to syntactic question-related features (e.g., [+Q], [+*wh*]; Adger & Svenonius, 2011), the context of use, or some sensitivity to the statistical differences in configuration type. In contrast, there was widespread unit sensitivity for statistical features. Lakretz et al. (2024) summarize: ‘[I]n contrast to orthographic and phonological features, we have not identified, in the present dataset, any single cell selective to specific values of higher-level linguistic features, such as syntactic ones’ (e.g., number, gender, tense, syntactic embedding, transitivity). Reporting this in the context of a dataset of 21 awake neurosurgical patients provides strong empirical support for ROSE. In Murphy (2024), a central prediction was that ‘single-unit recordings will *not* be able to detect effects of accessing a single linguistic feature and then utilizing it in multiple (underspecified) ways across linguistic sub-systems (e.g., syntax-form mappings).’ Specifically, the R level is composed of ‘ensembles of temporarily cooperating neurons’ that encode ‘bundles of features,’ specifically atomic ‘lexico-semantic features’ (Murphy, 2024; see; Collins, 2024) – pointing to a distributed code for syntactic features of the kind Lakretz et al. (2024) suspect is at play.

Călinescu et al. (2025) show that α/β power in EEG is modulated by lexico-semantic properties, but *not* by the syntactic mode of their composition; converging with the roles attributed to these rhythms in Murphy (2024). More recently, Chen et al. (2025) have used MEG to show that low

frequency activity reliably reflects the tracking of sentence-level syntactic features and the context-dependent parsing of word sequences.

Lastly, a recent review article systematically evaluated ROSE, VS-BIND (Calmus et al., 2020), Martin's compositional neural architecture (Martin, 2020), Ding's Hierarchical Structure Building hypothesis (Ding, 2020, 2023), and a number of other neural models of higher-order language. The authors concluded that 'ROSE's description of the syntactic parsing process is more comprehensive' than all other models (Qi et al., 2025). Qi et al. highlight that ROSE makes testable predictions for a broader number of frequency bands, and interactional dynamics, than other models. For example, VS-BIND focuses exclusively on $\theta - \gamma$ coupling (emphasizing hippocampus and dorsolateral prefrontal cortex); the DORA model and Ding's model focus on δ , paying less attention elsewhere; Bastiaansen and Hagoort (2015) segregate semantics/syntax purely into γ/β power dynamics; Nelson et al.'s (2017) parsing proposal focuses on γ (see Woolnough et al., 2023).

3. Structure meets statistics

I will discuss in this section some promising means through which cognitive neuroscientists might ground statistical processes alongside the symbolic forms of rule-based knowledge provided in ROSE. Afterwards, I will synthesize this discussion into an expanded ROSE model.

There are a number of ways we could embed probabilistic activation functions and frequentist, associationist mechanisms (LeCun et al., 2015; Pitkow & Angelaki, 2017) inside ROSE, such that symbolic representations become the *targets* of perceptual inference. There is a range of evidence that higher-order symbolic representations interact with statistical learning mechanisms (Gwilliams et al., 2018; Martin & Xu, 2018). Just as how models in the cognitive (neuro)science of language should always consider constraints on computation (van Rooij et al., 2019), they should also wrestle with limits placed on the neurobiology of perceptual inference (Friston, 2010; Murphy, 2020a; Tavano et al., 2022; Walsh et al., 2020).

3.1. The syntax–phonology interface

ROSE can set the stage for accounting for how syntactic instructions interface with phonology (e.g., intonation, stress, prosody; Frota & Vigário, 2018). Syntax often informs phonology through prosodic structures (Langus et al., 2012). These domains are influenced by syntactic structures but do not align with them in a one-

to-one manner (Samuels, 2011). Intonational boundaries are often determined by syntactic structure (e.g., 'Saul, who likes Mike, is a lawyer'). Certain types of syntactic information is accessible to phonology, but syntax is strictly phonology-free (Frota and Vigário 2018); Pullum and Zwicky 1988; Selkirk and Lee 2015).

For our purposes, I will merely highlight the likely neurobiological loci of these interacting levels of linguistic structure. Syntactic and prosodic information interacts via modulations in δ activity (Zeine & Poeppel, 2025), and local ensemble activity is highly sensitive to discrete dimensions of lexico-semantic and phonological information. I will not stake a claim here about which precise pieces of syntactic information (e.g., phases) need to be accessible to phonology, only that there is at a minimum some interface (Dobashy, 2003; Ishihara, 2007; Nevins, 2022; Newell & Piggott, 2011). Unlike syntactic tree structures, phonological representations consist of strings that lack generalized forms of recursion (Neeleman & van de Koot, 2006). Given this background, we can now turn to isolating more specifically how dimensions of symbolic and subsymbolic features interface.

3.2. Symbolic and predictive components

Phonetic information, surprise and sequence statistics can be recovered from single-unit recordings (Friston, 2025; Leonard et al., 2024), indicating a possible architecture whereby coordinated spiking activity is managed by high-frequency γ phase delivering surprise-based predictions (Gwilliams et al., 2022). Under ROSE, this integration of syntactic instructions feeding phonetic parsing would take the form of low-frequency δ/θ phase coupling (originating within frontotemporal language regions, whereby different syntactic features like nodal depth and dependency information provide distinct phonological instructions; Figure 3) with local high-frequency activity in auditory regions. Cross-frequency coupling provides a means to *integrate predictions* (Fiebelkorn et al., 2013; Grabot et al., 2019) across levels of linguistic information, with phase-based predictions being tested against amplitude-based sensorimotor transformations. Cross-frequency coupling, as implemented in ROSE, serves to *segregate* (Northoff & Huang, 2017) cortical computation into distinct but interfacing processing regimes (Mateos & Perez Velazquez, 2024), in contrast to equal frequency (1:1) connectivity that serves information integration. High γ power serves a fundamental binding role, as has long been hypothesized, but in combination with the cross-frequency coupling code for phrase structure (Murphy, 2024; Weissbart & Martin, 2024) this permits a connection between lower-order statistical inferences

encoded in γ to be mapped to a symbolic phase-coupled lower frequency code.

A feasible neurobiological mechanism can be found in how ROSE can actively and flexibly optimize phasal and amplitude-based parameters to become more directly attuned to syntactic error detection and reanalysis processes (Mandal et al., 2020), which are rendered at the spiking level. This doubtless involves, and triggers, processes of neuromodulation for the precision-weighting of prediction errors (Fu & Gagl, 2024; Möhring & Gläscher, 2023), and the modulation of ‘synfire’ chains (sequentially activated spiking patterns) for symbolic sequences (Zheng & Triesch, 2014). Reflecting the ever-flexible nature of symbolic knowledge, we do not need to assume that frequency bands here are strict types with rigid functional interpretations; rather, they are more likely to be what Martin (2020, p. 1423) calls ‘tokens of processes with physiological bounds that render them into functional types.’

Expanding on ROSE, I will provide two candidate mechanisms that can further ground this architecture, guided by considerations outlined in the previous work (Murphy, 2015b, 2020b). The first will be ephaptic coupling, and the second will revolve around dynamical motifs, both of which can guide the construction of minimally complex linguistic objects (i.e., clusters of features that assemble into morphemes) at the appropriate level of resolution whereby statistical and symbolic instructions will (presumably) naturally interface.

3.3. Minimal linguistic objects: ephaptic coupling

Spectral information like phase and oscillatory cycles offer a brain-intrinsic coordinate system by which the output of neurocognitive patterns can be interpreted (Giraud & Poeppel, 2012; Martorell, 2024; van Bree, 2024). The rich temporal dynamics of coupled oscillators (resonance, entrainment, reverberation, traveling waves, etc.) collectively expand a network’s ‘state space’ and enable more expressive, noise-tolerant computations (Singer & Effenberger, 2025). Adjacent to this notion of a phase code, I have discussed the potential role of ephaptic coupling (Murphy, 2015b) in an emerging neural infrastructure for language.

Ephaptic coupling, by influencing neural activity via extracellular fields (rather than via direct synaptic connections), provides one possible substrate for encoding probabilistic relationships, modulating signal-to-noise ratios, and shaping population-level dynamics (Miller et al., 2024). Groups of neurons may collectively encode stimuli features, with ephaptic coupling ensuring that the shared extracellular field conveys information about

the combined activity of these neurons (Anastassiou & Koch, 2015; Pinotsis & Miller, 2023). Emerging bioelectric fields can sculpt neural activity and aid the propagation of information across cortical sites (Pinotsis & Miller, 2023), making ephaptic coupling relevant to the mapping of lower-level statistical properties to network-level symbolic inferences. As Pinotsis and Miller (2023, p. 9878) put it, ‘the electric field enslaves neurons, not the other way around’ (for similar notions relating to ‘macro causal emergence’ that lie at the heart of ROSE, see Hoel et al., 2016; Potter & Mitchell, 2025). Low-frequency oscillations create rhythmic fluctuations in the extracellular electric field, which can phase-align membrane potentials and synchronize local ephaptic coupling. Oscillations can provide a temporal structure for updating predictions and errors (coded, for example, in Bayesian inference neural mechanisms; Sohn & Narain, 2021), with ephaptic coupling playing a role in local adjustments.

3.4. Minimal linguistic objects: dynamical motifs

Following on from some recent critiques of oscillatory models of language (Kazanina & Tavano, 2023a), I will take here what I see as a promising example of how to cash out my verbal formulations about population activity space and predictive processing in order to orient ROSE toward true neurobiological plausibility. To achieve this, I will consider *dynamical motifs*.

Driscoll et al. (2024) uncover in recurrent neural networks (RNNs) patterns of neural activity that implement specific computations through dynamics, such as ring attractors, decision boundaries and rotations, geared toward specific subtasks. This refers to the high-dimensional nonlinear dynamics around a fixed point skeleton that implement computations for a specified input (Durstewitz et al., 2023). Driscoll et al. (2024) offer a framework for understanding how a network can build up complex computations by reusing these ‘dynamical motifs’ – recurring, modular patterns in neural activity that are tied to specific computational subtasks. This modularity is evident when similar subtasks (like memory or decision-making) share fixed-point structures across different tasks (for similar directions, see Johnston & Fusi, 2023, 2024). This aligns with the discovery that representing domain knowledge through underlying relational structures (instead of purely statistical correlations) allows a model to generalize across domains more effectively (Doumas et al., 2022). Through a series of lesion experiments in their RNNs, Driscoll et al. (2024) show that dynamical motifs are a plausible candidate for being ‘a fundamental unit of compositional computation, intermediate between neuron and network.’

In the neurobiological space, ROSE seeks an analogous probing of minimal units in language. We can readily define a MERGE-based syntax as operating over distinct lexico-semantic trajectories. As such, a constellation of dynamical motifs, structured as an emergent phenomenon between units and networks (Miller et al., 2024), seem well poised to characterize some elementary features of language such as *commonly re-occurring* lexico-semantic feature spaces at the R/O levels, with RNNs being able to model some elementary components like feature bundling into a word-like element (Baroni, 2020). Phase-amplitude coupling in low-frequency oscillations could provide the timing framework that modulates when and how these stable representations are activated and combined. Beyond the level of lexico-semantic class, features specific to S and E levels (like the precise degree of syntactic complexity in a sentence, or the Gorn address of an ongoing incremental nodal inference in a tree structure) would be predicted to receive no directives from motifs.

Although the notion of dynamical motif is wholly based on artificial systems, the principles of optimality and robustness that Driscoll et al. (2024) discuss seem highly relevant to guiding biological theory formation (e.g., see Chen et al., 2021; Singer & Effenberger, 2025). Exploring these empirical predictions is becoming ever more feasible, thanks in part to developments such as MARBLE (Gosztolai et al., 2025), an approach that learns neural geometries flexibly, generalizes across contexts, and captures modular computations. These types of approaches entertain the idea that manifold geometry is a powerful inductive bias in learning dynamical representations, providing a potential clue to how real biological neural systems organize the learning of abstract representations. Meanwhile, in the psycholinguistic research space, new tools for annotating syntactic parsing processes onto speech acoustics (in ways that preserve the dimensionality of linguistic structure) enhance accessibility for experimentalists (Iaia & Tavano, 2024). Other tools relevant for modeling neural dynamics pertinent to ROSE are emerging (Jacobs et al., 2025; Mohammadi et al., 2024).

I provide a Table comparing dynamical motifs with other connectionist approaches (e.g., LISA, ACT-R, Holographic Reduced Representations, RAAM) in Supplementary Material. Empirical data for stable local attractors or ‘motifs’ in higher-order language regions remain limited – demonstrating them for lexical composition is non-trivial. In any event, the use of dynamical motifs for lexical information seems all the more plausible in the face of widespread, flexible and mixed high-dimensional coding across the cortical hierarchy (Posani et al., 2025). This space remains highly contentious in the

literature, in particular given that *the lack of firing* of neurons (the ‘syntax of silence’) can be just as causally effective as the firing of neurons (Pérez-Ortega et al., 2024).

In summary, I have related the recurring patterns of network activity in RNNs to the representational demands at the R and O levels. Note that in contrast to their suitability for learning lexico-semantic relations, in Supplementary Material I provide evidence that RNNs are likely inappropriate to faithfully model the S/E levels. Dynamical motifs can serve as the underlying modular substrate that could support compositional computation in artificial systems, providing some guidance for how mechanisms like spike-phase coupling can structure spiking codes for distinct, emergent lexico-semantic classes that occupy varying trajectories in conceptual space. These considerations for the R and O levels are in line with the general idea that the brain organizes many semantic concepts along smooth, low-dimensional manifolds (Zhang et al., 2025) that preserve relationships between concepts and lexical elements. Tying dynamical motifs to more global, phase-coded syntactic signals provides an articulation of neurosymbolic principles. Naturally, a topic of current research concerns how these ‘emergent attractors’ actually anchor to local synaptic or dendritic processes in real cortical circuits. One possible avenue of investigation involves exploiting spike-phase coupling indices as functional readouts of when motifs form and dissolve.

3.5. Representing a minimalist grammar parser

As presented in Murphy (2024), ROSE provided a biophysical model of oscillatory coupling in the absence of a more temporally precise algorithmic settlement for parsing. Here, I will demonstrate how ROSE can provide the infrastructure for a minimalist grammar parser, allowing for a more restrictive and precise set of experimental predictions.

Essentially, I will assume that the phase offset of each low-frequency cycle instructs local circuits about the type of syntactic parsing action to be executed, whereby the network dynamically modulates local membrane potentials as a function of the instantaneous phase of the wave. Specific phase offsets will thereby ‘endorse’ particular parsing options over others. Concretely, we can assume that Shift (lexical retrieval) operations occur during low-frequency phase troughs, when the system is maximally receptive to bottom-up input, while Reduce (collapsing constituent structures) operations are more likely to occur during phase peaks (Figure 5). This gating function is how the symbolic top-down parse can instruct bottom-up lexical selection

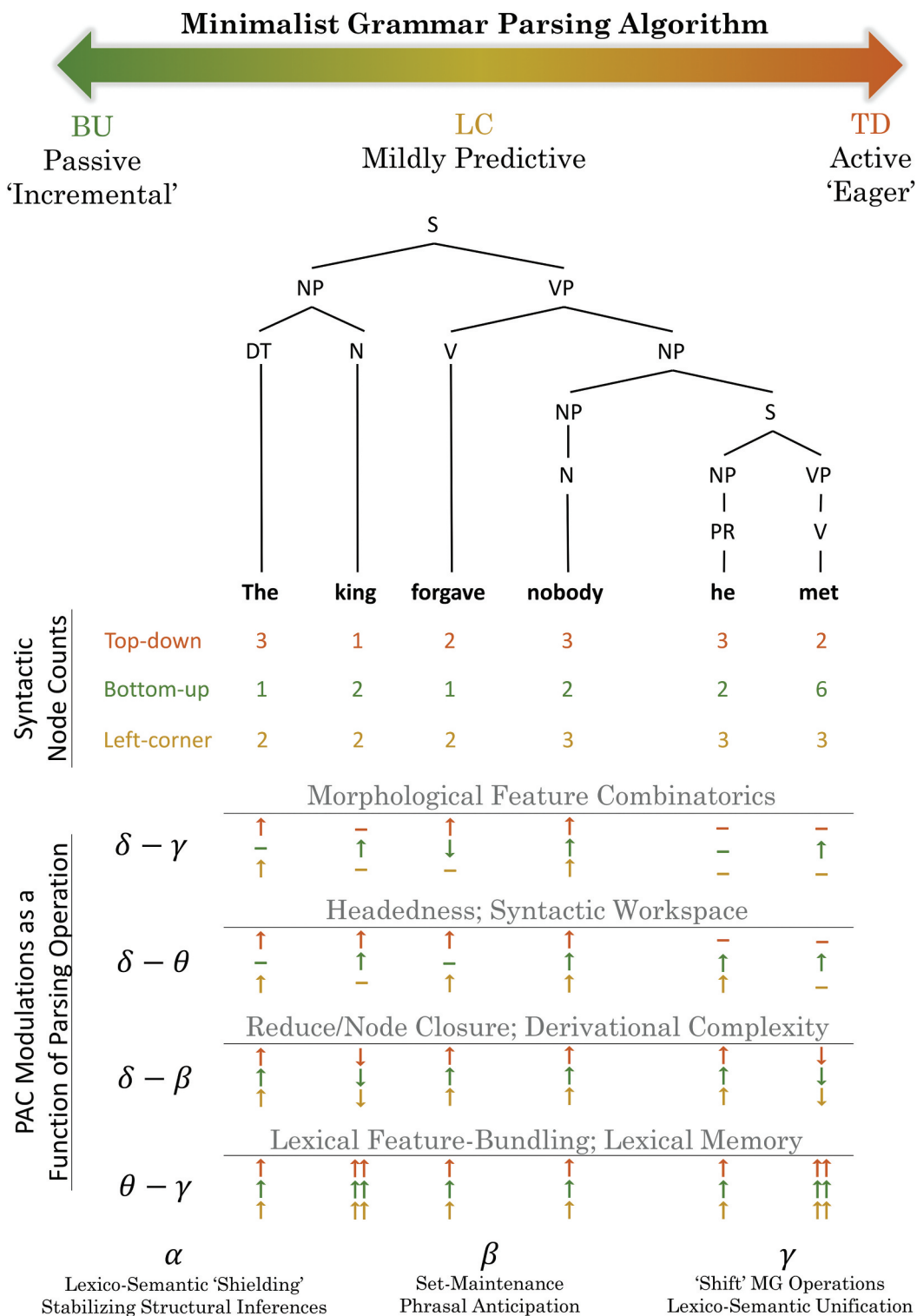


Figure 5. A tree structure with node counts for minimalist grammar parsers. A representation of different parsing counts for open nodes under minimalist grammar parsers, which afford different goal-driven inferences across a spectrum of predictive 'eagerness.' Under the tree, these word-by-word counts represent the number of parser actions taken before moving to the next word. Under ROSE, low-frequency oscillations act as phase codes for symbolic operations, while higher-frequency activity (γ) encodes mostly fast, statistical, sensory and conceptual information. One can hypothesize that each discrete operation in a left-corner parser – say, Shift (lexical retrieval) or Reduce (merging the left corner with a higher constituent) – is tied to a particular phase in a slow oscillatory cycle. Below the tree, example hypothesized increases, decreases or stable signatures (–) are linked via their color to parsing strategies based on their principal (though not necessarily exclusive) oscillatory drivers. This representation compresses some subtleties mentioned in the main text; e.g., although all three models follow a similar overall pattern, note that left-corner parsing may show more gradual changes in $\delta - \beta$ coupling in practice. Parsing models would differ slightly on their predictions for the O level with spike-phase coupling and γ in terms of *when* elements are predicted to be assembled. For details about α/β see reviews in Murphy (2024; Meyer (2018); Grabenhorst et al. (2025).

and thus shape which item is integrated next. At each step of parsing, one of the following actions is selected:

GEN (Generate): Generate a terminal symbol.

NT (Non-terminal): Open a nonterminal symbol.

REDUCE (Merge/Label): Close a nonterminal symbol that was opened by NT.

SHIFT (Lexical access): Takes the next lexical item (a terminal) from the input and pushes it onto the parser's stack.

I will take as a plausible candidate parser for human sentence processing a 'left-corner' minimalist grammar parser (Hunter, 2019; Li et al., 2022; Liu, 2024). A left-corner parser is an incremental strategy that begins parsing from the leftmost incoming element and immediately integrates it as far as possible into the partial tree before proceeding, and typically leads to less backtracking. MERGE (Reduce) would be applied as soon as a valid combination appears (at the 'left corner' of a parsing rule), rather than waiting to build structure bottom-up (Figure 5). Left-corner parsing combines top-down and bottom-up information: The top-down information is the syntactic category being searched for, and the bottom-up information is the single element being processed at a given period (Johnson-Laird, 1983). Left-corner parsers do not need to decide on all the positions in which a filler phrase checks features before that phrase can be scanned, hence is more incremental and amenable to accommodating 'Late Closure' and active gap-filling effects (W.-Y. Chow & Zhou, 2018; Omaki et al., 2015).

A representation of different counts for open syntactic nodes based on minimalist grammar type is provided (Figure 5). In this example, all parsing models predict an increase in $\delta - \theta$ and $\delta - \gamma$ PAC across frontotemporal language sites at Word 4, but they differ on the final word. In terms of traveling wave dynamics, sensitive to workspace demands, the left-corner model predicts a steady propagation from posterior temporal to anterior temporal and frontal regions across the sentence, with a marked increase in the number of sites the low-frequency wave is coupled to at Word 4. Multiple waves would be recruited for deeper nested structures but would critically be phase-aligned with the 'initiating' wave, demonstrating true hierarchical expansions rather than repeated serial chunkings. Indeed, multiple waves would be hypothesized to be superimposed across different cortical routes, depending on workspace and task demands. I refer the reader here to Mohan et al. (2024) for intracranial evidence that the direction of frontotemporal δ and θ traveling waves (specifically, 2–13 Hz) modulates memory processing (with a posterior-to-

anterior trajectory during encoding, as predicted under ROSE; Murphy, 2024). Since syntactic structures have no maximum duration limit (driven by syntactic recursion), future research should explore how traveling wave dynamics 'cope' with increasingly longer, and deeper, discourse units.

Simulation work in particular will be needed here. As part of an ongoing effort and work-in-progress, I have released some initial modeling code for certain components of ROSE (e.g., establishing headedness via $\delta - \theta$ dynamics; lexical assembly via $\theta - \gamma$ coupling; generic combinatorics via $\delta - \gamma$ coupling; itinerant dynamics of the E level; left-corner PAC dynamics; dynamical motif emergence) under 'ROSE-Sim' (github.com/ElliotMurphy91/ROSE). This can facilitate analyses of PAC routines, motif-RNN hidden states, and traveling wave itinerant dynamics. ROSE-Sim provides a skeleton from which further experimental modeling can be easily conducted, at least with respect to some basic predictions of ROSE.

The present perspective offers an alternative framework for how Ding (2020, 2023) accommodates workspace dynamics by focusing mostly on δ , although it remains unclear in the literature what the exact scope of syntactic working memory is (Townsend & Bever, 2001), how 'large' it can be (see Ding, 2025 for discussion of the neurobiology of sequence chunking), how item order is encoded (Liebe et al., 2025), and how formal accounts from linguistics can be properly utilized (e.g., the hypothesis that syntactic workspaces are 'binary forests,' or finite disjoint unions of abstract binary rooted trees; Marcolli et al., 2025). Some insights come from the discovery that longer grammatical dependencies are linked to processing difficulty (due to working memory demands) *regardless* of whether these dependencies are nested or cross-serial (Yadav et al., 2025), narrowing the likely candidates that will impact E level dynamics (Gibson, 1998).

In terms of timing, we would expect Reduce modulations of PAC dynamics to onset immediately after lexical access (which is typically around 180–250 ms after word onset), at approximately the 250–400 ms window (Murphy, Rollo, et al., 2024; Pykkänen, 2019; Woolnough et al., 2023), aligning with the introduction of a new δ cycle. Spike-phase coupling would trigger appropriate dynamical motifs to generate feature-bundles associated with a given word (see also Xu et al., 2024); a local excitatory-inhibitory loop stabilizes the motif; a phase offset or amplitude modulation signals completion; and a traveling wave reset moves the partial parse forward. At the R/O levels, encountering 'the' and 'king' (Figure 5) would lead to binding lexico-semantic features via circular convolution (Calmus et al.,

2020; Mitchell & Lapata, 2010; Plate, 2003) or vector-derived transformation binding (Gosmann & Eliasmith, 2021), while information about the dominating nodes would be instantiated via $\delta - \theta$ PAC across frontotemporal cortex. To give another simple example of a filler-gap dependency, spike-phase coupling could transiently re-instantiate the lexical features of the filler when the parser hits the gap position.

One point of clarification here pertains to my argument that the parsing steps and spatiotemporal dynamics I invoke here (Figure 5) are to be thought of as *the principal but not exclusive drivers of parsing operations*. For example, when I argue that $\theta - \gamma$ coupling provides a readout of lexical feature-bundling, I am not committed to the claim that *only* this PAC relation subserves this computation – rather, it is to be thought of as the relation with the largest explanatory-causal scope for this parsing operation, and the one that best predicts and drives it. It may be that a many-to-many mapping is needed to derive headedness and lexical feature-bundling, not just one type of PAC relation, but for now I assume these to be critical drivers of the relevant computations. Of course, this still leads to direct testability, but with the important notion of necessary versus sufficient features in the backdrop.

In connected work, Sugimoto et al. (2024) show that hierarchical models like recurrent neural network grammars (RNNs) better explain human brain activity over frontotemporal language sites in fMRI than sequential models. RNNs jointly generate word sequences and their syntactic structures (Dyer et al., 2016), and outperform sequential models in predicting syntactic dependencies (Wilcox et al., 2019) and eye movements (Yoshida et al., 2021). Brennan et al. (2020) used RNNs with EEG to show that around 200–300 ms after encountering a word explicit hierarchical composition seems to occur.

Minimalist grammar parsers (in particular left-corner RNNs) therefore provide a psycholinguistically plausible model for core elements of syntactic parsing (see also Chowdhury & Caragea, 2021). We can assume here that the stable fixed points of dynamical motifs correspond to the intrinsic representations of words, while the shifts in attractor landscape reflect the outside modulation by low-frequency phase signals that bind words into coherent phrases. This architecture is not unlike Stewart et al.'s (2014) computational model for left-corner parsers, but less tied purely to models of spiking neurons.

Left-corner parsing seems to be a strong candidate for the brain's 'default' parsing strategy, with different contexts potentially triggering a shift toward more or less predictive strategies (Slaats et al., 2024). Importantly, ROSE is well suited to capture this flexibility – it is *not* inherently tied to any particular minimalist grammar

parser. Indeed, language production seems to recruit top-down parsing strategies and comprehension more bottom-up strategies (Giglio et al., 2024). As already anticipated in the literature (Brennan et al., 2020; Giglio et al., 2024; Murphy, 2024), different brain regions might encode distinct parsing strategies.

The present approach departs from others in the field that focus less on the format and structure of linguistic computation and place greater emphasis on domain-general processing. Construction Grammar and Usage-Based accounts place heavy emphasis on performance factors like the relevance, appropriateness, frequency and priming of a given construction. Theories are anchored highly around performance constraints like noise, competition, and time pressure. ROSE can certainly accommodate these and other important performance constraints (see Section 3.8), but it begins with the assumption that if we want to explore the neurobiology of language we should seek a highly structured and formal characterization of what language is.

3.6. Comparison with other accounts

The predictions in Figure 5 emerge from a more psychologically plausible set of hypotheses for compositional syntax – and a stronger attempt to incorporate specific parsing algorithms – than prior frameworks:

- (1) ROSE supports Meyer's (2018) view that one of the roles of high γ is to index 'diagnosis of lexico-semantic contextual fit' (though this is not the only role; Murphy, 2024), and his views about α indexing some aspect of lexico-syntactic working memory (Murphy, 2015b). However, the oscillatory model in Meyer (2018) focuses on 'chunking,' 'storage' and 'retrieval' (and 'capacity restrictions' that emerge), which is not a candidate model for natural language syntax (see Kazanina & Tavano, 2023c). Still, a potential strength of Meyer's framework over mine is his finer-grained integration of numerous phonological processes and how they aid syntactic processing.
- (2) Certain vector-symbolic models like VS-BIND (Calmus et al., 2020) provide highly plausible and elegant accounts of ordered sequential chunking but in a manner somewhat adjacent to classically compositional MERGE-based hierarchical syntax (Figure 2).
- (3) As mentioned, the models in Murphy (2015b, 2016, 2020b) do not invoke many parsing considerations.
- (4) The model in Martin (2020) predicts low-frequency power modulations in response to

structure-building, although with fewer finer-grained computational-level predictions, making it more difficult to falsify. ROSE (Figures 1, 5, 6) does not simply predict ramping of low-frequency power during composition. Nevertheless, a potential benefit of Martin's model over mine lies in its intrinsically *integrative* framework for relating distinct cognitive faculties (memory, retrieval, semantic composition, sensory sampling), whereas ROSE places more of an emphasis of language-specific computations and representations (Figures 1–3, 5) (part of my motivation for this concerns how MERGE seems to be optimized for things like 'theta theory' (Marcolli & Larson, 2025) and how node closures are achieved via a proprietary algorithm (see Section 2.2), but this topic takes us further afield). Yet, this may potentially render Martin's conception of the 'language network' more opaque, being intrinsically linked to memory and perceptual systems, which may not be clinically tractable or neurobiologically plausible (see citations in Section 2.3). Regardless, Martin's model is certainly plausible from an evolutionary perspective, as discussed in Murphy (2020a) (see also Murphy, 2019).

- (5) The present model differs in predictions from Boeckx and Theofanopoulou (2018), for whom lexicalization is realized via $\alpha - \gamma$ coupling, labeling via β , and workspace maintenance via $\theta - \gamma$. Other models (e.g., Cross et al., 2018) do not sufficiently differ from prior proposals (Murphy, 2015b, 2016) to deliver novel predictions.
- (6) The oscillatory framework outlined here may prove to be directly linked to an emerging consensus (Delogu et al., 2025) that the N400 event-related potential component indexes *access* (or θ/α -coordinated and high frequency components of ROSE), and the P600 indexes *integration* (elements relating to δ).
- (7) The present model is in contrast to the framework for isolating language in Shain and Fedorenko (2025). Nevertheless, one of the methodological advantages of Shain and Fedorenko's work is their language-selective network's highly robust isolation from many major non-linguistic cortical networks (music, vision, theory of mind), providing a useful cortical framework for subsequent exploration from a more computationally explicit parsing approach.

3.7. Universal Neural rammar

In the architecture that emerges here (Figure 6), a number of subsymbolic information sources can be

represented at R: For example, lexical frequency (spiking rate modulation), contextual predictability (amplitude variations reflecting prior probabilities or surprise), and lexical associations (patterns of co-activation among spiking populations). R/O levels provide the minimal morphemic units for what we can then model with an RNN for a specific parsing strategy, while RNNs can model how dynamical motifs emerge for different lexico-semantic spaces. This infrastructure can provide a tentative foundation for what we can think of as a 'Universal Neural Grammar' – the species-specific neurocomputational architecture for syntax. The stability of the generalized phase code is here contrasted with the diversity and heterogeneity intrinsic to multi-modal sensorimotor transformations (Figure 6). ROSE can be seen here as offering a 'mesoscopic protectorate' for natural language (Laughlin et al., 2000), whereby syntax is a complex emergent phenomenon that cannot be approached sufficiently by looking under-the-hood purely at the R and O levels of neurobiological organization.

Iaia and Tavano (2024) argue convincingly that 'if syntactic categories cannot be defined based on temporal statistics, then any homologous mapping of syntactic structure onto a neural time scale, for example such as is provided by neural oscillations, is bound to fail.' It is certainly true that syntactic categories (like Noun Phrase) are not bounded by temporal statistics. 'I' can be a phoneme, syllable, word, Noun Phrase, or discourse utterance. What ROSE aims to capture is not just the 'temporal unfolding' of parsing (although it can accommodate this accurately; Figure 5) but more importantly the neurobiological organizational scales that generate endogenous inferences about syntax. The mechanisms of ROSE *respect set grouping* (non-associativity). It is within the richness of phase-coupled embeddings that syntactic categories emerge; they are here incorporated within the oscillatory *configuration* rather than emerging at a specific *duration*. For instance, the 'Structure' (S) level carries out hierarchical syntactic inferences via low-frequency oscillatory codes *not* because 2 Hz or 3 Hz directly defines a certain category, but because the brain must implement symbolic computations somewhere in neural time. The brain exploits certain timescales (and cross-frequency coupling) to build those categories online and in a robust, error-tolerant way.

As mentioned, it is becoming increasingly apparent (e.g., Coopmans et al., 2025; Giglio et al., 2024) that theoretical neurolinguists do not need to simply seek a plausible infrastructure for one specific type of parsing algorithm, but multiple possible strategies. I have argued in this section that ROSE furnishes the

• *Universal Neural Grammar* •

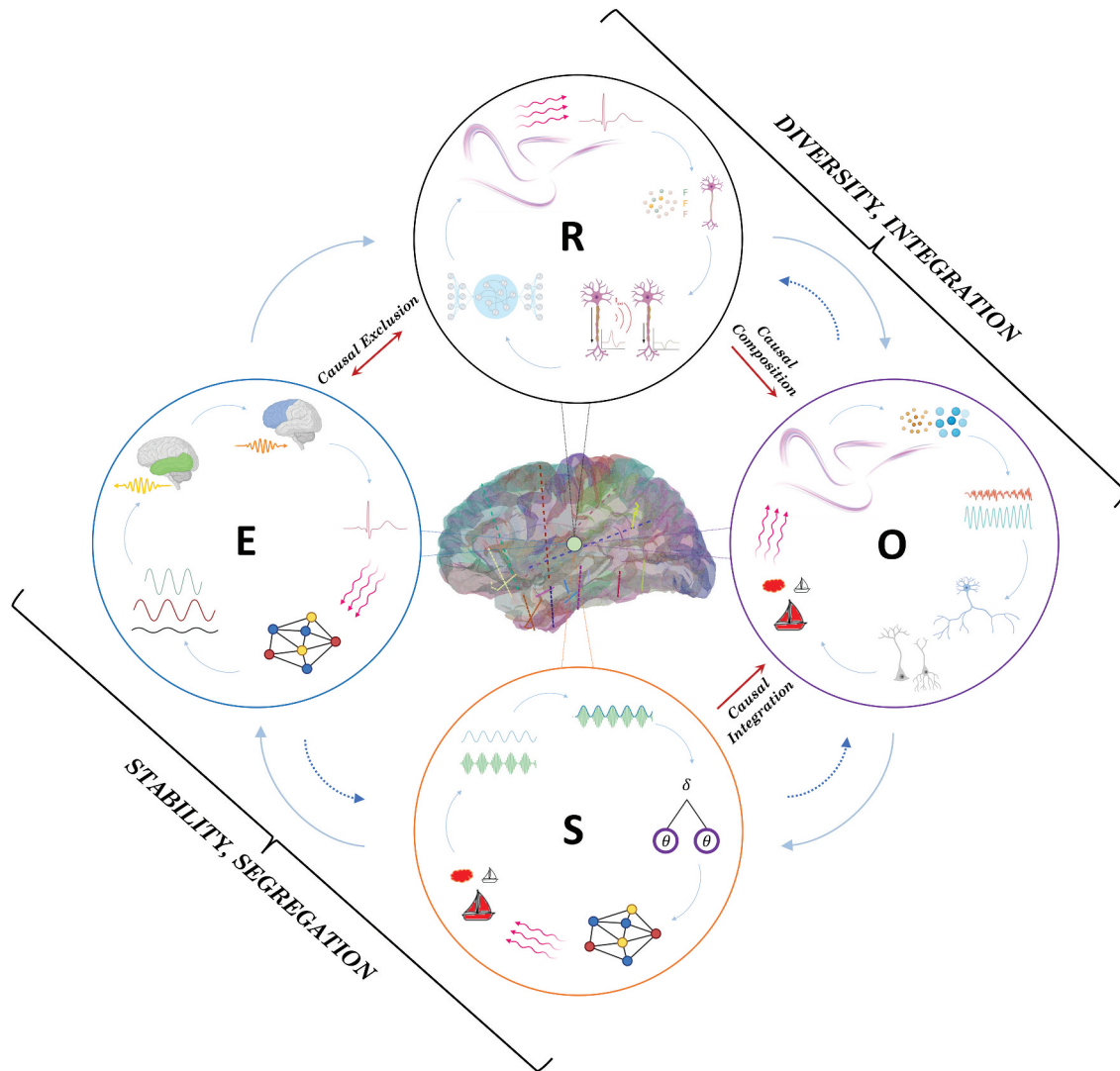


Figure 6. ROSE as a ‘mesoscopic protectorate’ and a hybrid symbolic-connectionist architecture. Proposed foundations for a ‘Universal Neural Grammar.’ Blue arrows connecting circular units represent information integration; prediction over levels is represented by dotted arrows. Centre: cartoon depiction of an example stereotactic-EEG (sEEG) electrode placement in humans. Each of the four surrounding units represent the components of ROSE with different causal properties (E-to-R: causal border excluding R from directly causing shifts in E; R-to-S: parts at R have cause–effect relations with S; S-to-O: causal irreducibility of the whole to its parts) (Hoel et al., 2016). Within components, the three red arrows depict the completion of the component’s computational contribution and the point of accessibility to the immediately higher level (for the E component, the red arrows instead indicate the point of influence back down to R, i.e., when low-frequency traveling waves drive phase-resetting). ‘Representation’ (R) component (beginning at symbol immediately to the right of the red arrows): spiking shapes responsible for coding various statistical features cluster into ensembles of localized representations; a depiction of ephaptic coupling (I_{ext} represents an externally applied current or electrical field); a representation of an RNN able to model lexico-semantic feature-bundling across both statistical and symbolic feature types; a set of trajectories that emerge from the combination of spiking shapes (some influenced by ephaptic coupling) that provide the foundations of dynamical motifs (purple trajectories). In terms of possible neurobiological implementation for dynamical motifs, within pMTG and IFG, based on well-known physiological properties recurrent excitatory loops in L2/3 could sustain attractor motifs, L1 inputs could provide phase resets, and L5 outputs could coordinate higher-level traveling waves. ‘Operation’ (O) component: dynamical motifs emerge from R and spike-phase coupling to synchronize bundles of features; features relevant to lexico-thematic and lexico-syntactic operations are bundled and coordinated via spike-phase coupling, e.g., intersective adjective-noun combinations like ‘red boat.’ ‘Structure’ (S) component: a depiction of the various possibilities for phase-amplitude coupling that emerge from accessing the γ feature-bundles provided by O; PAC operations (see Figure 5); a restricted path for subsequent parsing options (represented here by colored nodes). ‘Encoding’ (E) component: the set of structures outputted by S are transferred across frontotemporal language network nodes via traveling waves. The wave front then either (i) resets or (ii) continues traveling to additional cortical sites (e.g., parietal loops) for extended memory/prediction operations. Traveling waves can physically distribute the same S-level parsing state across multiple cortical patches before the next element of structure is incrementally composed. See also ‘ROSE-Sim’ (github.com/ElliottMurphy91/ROSE) discussed in the main text.

appropriate levels of *temporal flexibility* and *mechanistic stability* needed for this. A flexible adaptation to different parsing strategies (Figure 5) algorithmically guiding a MERGE-based syntax (Figure 2) would be permitted based on whichever neural regional hub (Figure 4) drives initial symbolic inference generation (Figure 3), guided by contextual concerns of use (comprehension; production; task demands) (Figure 6).

For instance, in everyday cases the use of statistical inferences may often suffice to extract a type of shallow parse (a ‘good-enough’ parse) for linguistic structure (Frances, 2024). There is considerable individual variation in the extent to which comprehenders engage in syntactic prediction (Chow & Chen, 2020; Ferreira & Qiu, 2021; see also Chen et al., 2025). Every case of semantic (next-word) prediction logically involves a syntactic prediction, since all words belong to a grammatical class (Matchin et al., 2017). At the same time, many predictions can be successful simply by virtue of lexico-semantic associations (Jackendoff & Audring, 2020), although semantic prediction mechanisms can also prime words that may *not* fit the correct syntactic category (van der Burght et al., 2021). ROSE therefore needs to flexibly adapt to how individuals access and exploit the resources of their ‘I-language’ (Chomsky, 1986, p. 22).

3.8. A hybrid neural code for syntax?

My treatment of the neural code for compositional syntax follows the spirit of many neurosymbolic approaches in artificial intelligence and cognitive science research (Marcus 2001; Garnelo and Shanahan 2019; Edelman 2008; Alers-Valentín et al. 2023 Dhanraj and Eliasmith 2025; Wermter and Lehnert 1989; Eliasmith et al. 2012; Garcez et al. 2015; Phillips 2020; Baggio and Martin 2020; Minsky 1990; Nilsson 2007; Graves et al. 2016; Smolensky et al. 2022), and it is (vaguely) in the spirit of the ‘physical symbol system hypothesis’ (Newell & Simon, 1976; Nilsson, 2007). Assessing symbolic versus associationist models of syntax requires going beyond viewing syntax as a monolithic computational regime, as is common in the field. Due to the fact that structural and sequential properties of language very often correlate (Takahashi & Lidz, 2008), with transitional probabilities being a decently useful cue for phrase structure (though, infamously, not always; Chomsky, 1957; Slaats & Martin, 2023), it has proven difficult to confidently isolate them in neurobiology.

We can now negotiate into appropriate levels of ROSE (a model that entertains both periodic and aperiodic components) the necessary aspects of probabilistic reasoning, as shown in recent literature (Dekydspotter et al. 2024; Shi, 2024; Weissbart & Martin, 2024; Yeaton, 2025; Yu et al., 2024). In addition to acknowledging that

statistical information serves to cue appropriate symbolic inferences (Hale, 2003; Karimi et al., 2024; Martin, 2020; Takahashi & Lidz, 2008), we can invoke the following assumptions:

- (1) Oscillatory mechanisms facilitate temporal integration of statistical patterns into a phase-coupled top-down symbolic code (Luo & Ding, 2024).
- (2) Distributed representations in a connectionist architecture for lexical learning are coordinated over time by frontotemporal traveling waves (Weissbart & Martin, 2024). $\delta - \beta$ phase-amplitude coupling in posterior temporal cortex is jointly sensitive to lexical surprisal and syntactic complexity (tree depth); consider also the sensitivity of β power to syntactic prediction (Murphy et al. 2022; Zioga et al. 2022). $\theta - \gamma$ phase-amplitude coupling is jointly sensitive to syntactic node closures (in posterior temporal cortex) and word-level entropy (in inferior frontal cortex), pointing to a feasible interface between symbolic structural inferences being cued by lexical statistics (Weissbart & Martin, 2024).
- (3) Interactions between lateral prefrontal and posterior middle temporal cortical sites with various cortico-subcortical loops achieve integration between symbolic and probabilistic knowledge (Murphy et al. 2022).
- (4) Top-down predictions about higher-order structure imposed by ROSE could be implemented via firing rate functions, non-linear postsynaptic responses, and modulations in neuronal connectivity (Friston et al., 2017), buttressed by the hypothesis that overlapping patterns of population activity encode statistics that summarize both estimates and uncertainties about latent variables (Pitkow & Angelaki, 2017).
- (5) Properties of synaptic computation (Aitchison et al., 2021; Soltani & Wang, 2010) would here execute probabilistic inference for lexico-semantic feature spaces, a promising candidate for a biophysical instantiation of the Bayesian decision rule (e.g., more uncertainty leading to greater variability in postsynaptic potential size). This Bayesian coding can be seen at the level of single cells (Friston, 2025), circuits (Narain et al., 2018) and populations (Dubreuil et al., 2022; Ganguli & Simoncelli, 2014) levels.

One of the core architectural components of ROSE, phase-amplitude coupling, can: (1) interface with frequency-specific statistical encoding; (2) ensure that

local lexical statistics are aligned with global phrase-level predictions; (3) provide probabilistic representations via phase encoding; (4) encode statistical interactions (e.g., joint probabilities); (5) and represent statistical gradients (e.g., shifting probabilities or contextual relevance). In terms of basic predictions for error encoding, subsymbolic errors arise when local spiking ensembles mismatch the expected distribution, whereas complex syntactic errors might show up in the low-frequency phase code as an attempt to MERGE items that yield illicit labeling (e.g., a deterministic feature mismatch).

Under ROSE, synaptic weights would rapidly change to capture statistical dependencies encountered in data (Friston, 2025), but those weights would still be under the purview of spike-phase coupling which, via ROSE, is functionally coupled with low-frequency phase codes indexing structural/hierarchical inferences (Chen et al., 2025). Different spiking shapes could then be responsible for coding various statistical features of localized representations (Leonard et al., 2024; see also Wang et al. 2025b). Spike-phase coupling is a strong candidate for implementing neural spiking precision to syntactically-driven predictions, with γ being a stronger candidate for phase coordination in most cases than the LFP (Leonard et al. 2023). For now it seems clear that high frequency γ power can encode both statistical and symbolic properties, as in the case of both melodic and syntactic representations (McCarty et al., 2023).

My approach here is similar in spirit to Baroni et al.'s (2014) pioneering approach whereby certain aspects of word meanings are represented as co-occurrence vectors (like in distributional semantics) but compositionality is delivered via a syntax-driven calculus of function application, from symbolic principles of formal semantics. My approach is also highly sympathetic to analysis-by-synthesis approaches to linguistic working memory (Bever & Poeppel, 2010; Townsend & Bever, 2001) that integrate associative and symbolic information during parsing.

Cross-frequency coupling allows symbolic rules to flexibly shape connectionist representations at the R/O levels, creating a contextualized hybrid model (Figure 6). This mirrors how symbolic AI systems impose rules on a neural network's learned patterns (Marcus, 2001). Under ROSE and the postulated interfaces with probabilistic mechanisms, symbolic representations and rules are implicitly embedded in phase dynamics. Also note that the major constraints on MERGE (Chomsky et al., 2023) are here not localized uniformly, but across levels (e.g., lexico-thematic constraints during 'Shift' at R/O; labeling constraints during 'Reduce' at S/E), leading to clear paths for testability.

In summary, ROSE (Figure 6) posits that the brain orchestrates complex cognitive functions through a hierarchy of dependencies:

3.8.1. Global scaffolding

Traveling waves set global timing for lower levels of ROSE, providing a continuously shifting phase map over the cortex. This keeps track of the current parsing state and workspace.

3.8.2. Local scaffolding

Spike-phase coupling and ephaptic coupling synchronize local networks, complementing the global function of traveling waves. As the traveling wave propagates, its associated low-frequency field modulates the membrane potentials across local neural clusters. This field effect reduces phase noise and reinforces the alignment of spike timing across neurons.

3.8.3. Symbolic structure

Phase-amplitude coupling drives γ bursts that instantiate dynamical motifs in local circuits (i.e., stable, recurring patterns of activity that act as computational primitives for assembling commonly bundled lexico-semantic features), while coupling between lower frequencies endorses when parsing operations can be optimally executed.

3.8.4. Sequential structure

Spike-phase coupling defines discrete computational windows for lexico-semantic features to be jointly activated and allows for the checking of predictions against high-frequency sensorimotor transformations.

It is my contention, then, that the neural code for syntax will be a hybrid one. ROSE provides a tentative foundation.

3.9. Neural mechanisms for ROSE

Lastly, what can be said about the causal-mechanistic basis of the elements of ROSE outlined here? Picking up on some earlier discussions (see Murphy, 2015b, 2020b, for details), we can move from molecular biophysics up to inter-areal field dynamics, showing how each proposed phenomenon (spike-phase coupling, $\delta - \theta - \gamma$ PAC, β/α workspace signals, traveling waves, ephaptic fields, dynamical motifs) can arise from known circuitry. This relies not on a single-site mechanism, but a stack of nested causal processes – forming a mesoscopic protectorate for ROSE. That said, I should note that some of the framing in this section below remains speculative, and it may be useful for future empirical work to more carefully investigate plausible physical infrastructures for ROSE. My intention here is to offer some reasonable grounding for a relatively abstract neural code.

For the R level (*local feature bundles, single unit codes*), at the dendrite scale we likely have NMDA-spike clusters on apical tufts (major excitatory input), gated by GABA_A from chandelier and neurogliaform cells. This lets multiple lexico-semantic features co-activate yet remain isolated until a θ -phase gate opens. Multiple lexico-semantic features can depolarize simultaneously yet stay ‘quarantined’ until the larger θ gate opens. At the minicolumn scale, we have a recurrent L2/3 pyramidal loop with fast PV⁺ feedback delivering γ bursts, and SOM⁺ dendritic inhibition yielding a θ envelope that modulates γ amplitude (Maier et al., 2010). This will later generate the $\theta - \gamma$ code that assembles morphemic feature bundles, with the strength of γ reflecting feature salience. At the ephaptic micro-field scale, we would have out-of-phase γ volleys creating sub-millivolt extracellular gradients that bias neighboring tufts. This implements probabilistic priming among nearby lexical candidates without synapses. When γ bursts entrain local PV interneurons, the ensemble becomes an R-level token accessible to O-level binding.

For the O level (*feature bundling, sensorimotor transformations, motif formation*), at the scale of spike-phase coupling, spikes from L2/3 pyramidal cells are locked to the phase of local γ , while θ plays an envelope/gating role. PV-interneurons set the γ carrier. θ amplitude modulates γ power but spikes remain phase-locked to γ , and the feature bundle with the largest spike- γ phase-locking value becomes most salient/prominent for interpretation. If we aim to respect commutativity, each feature’s spikes can still lock to some γ phase angle, but that angle is treated as a hash key (an arbitrary slot in the 0–360° cycle) rather than as a serial order marker. Likewise, θ phase could modulate the magnitude of every γ burst simultaneously; it may not impose an ordinal sweep across features. Most of my focus here has been on exploring the non-associativity aspect of MERGE, but these concerns over commutativity could also be implemented in ROSE (indeed, for headedness at S the algorithm is concerned with PAC strength, so its outcome already respects commutativity). The brain may not build full tree-structures during parsing, but, rather, online parsing may be constrained by abstract syntactic cues, though in a chunked, probabilistic, and resource-bounded fashion. Lastly, concerning dynamical motifs, we can invoke recurrent E-I microcircuits running through a fixed-point ring attractor (see Figure 6 description).

For the S level (*MERGE-specific criteria*), δ originates in deep L5 IT neurons, θ in L2/3 SOM⁺ interneuron networks, and γ from PV-mediated loops. It may be possible to invoke here phase precession: Within each δ cycle, θ troughs precess $\sim 90^\circ$ across nested nodes, giving

a temporal order identical to the syntactic hierarchy. This may be another way to implement non-associativity ($((\alpha\beta) \gamma) \neq (\alpha (\beta \gamma))$): different inference paths occupy different phase offsets. For the β -sensitive node depth count (Figure 5), a re-entrant cortico-striato-thalamo-cortical loop accumulates β power linearly with open constituency depth, providing a continuous node count variable.

For the E level (*workspace capacity, syntactic broadcasting*), at the traveling wave scale we have delay-coupled L5 pyramidal sheets linked by cortico-cortical U-fibers, which with slow conduction produces posterior-to-anterior phase gradients. This physically ships the completed low-frequency complex off to relevant association cortices for further access and inference. For other aspects of low-frequency workspace limits, we have frontoparietal α bursts from pulvinar suppressing additional δ initiations once ~ 4 –5 complexes coexist (depending naturally on other relevant memory factors) (Murphy, Hoshi, et al., 2022). With respect to cross-scale integration, we might have inferior frontal ensembles computing syntactic anticipation by feeding predictions back (via thalamus and superficial feedback pathways) to posterior temporal tufts, which bias upcoming θ windows.

These comments help further refine the code for labeling in greater detail. Labeling is effectively realized in three microscale steps: First, a PAC-sensitive interneuron pool in posterior MTG/IFG selects the θ burst that shows the maximal δ – θ modulation index, thereby electing a single head. Because each spiking assembly already carries a category vector (θ PHASE OFFSET, γ SUB-BAND), the ‘winning’ burst inherits an unambiguous class tag (e.g., NP = $\gamma@0^\circ \pm 30^\circ$, VP = $\gamma@120^\circ \pm 30^\circ$), with featural properties of γ contributing to the projection of the phrasal category (as discussed above) (Artoni et al., 2020). A brief β ‘commit’ burst (Lundqvist et al., 2024) (functionally distinct from the δ – β nodal depth tracking complex) then propagates this vector to a fronto-temporal label register while PV feedback suppresses the daughter γ carriers. Subsequent MERGE gates consult the register’s read-only vector-to-category map, so previously merged items cannot be re-opened. Effectively, phase indicates *where* in the tree-structure, γ sub-band properties indicate *what* kind of semantic properties the phrase has (respecting the division of labor across R/O and S/E). Notice that γ power still scales with salience and contextual prominence. For example, two adjectives merged with a noun share the same δ trough; their θ bursts differ only in γ sub-band properties, maintaining order-insensitivity while still separable by downstream decoders. For deeply center-

embedded phrases we predict a cascade of $\delta \leftrightarrow \theta$ cross-resets that stack these registers hierarchically. Thus, the algebraic structure of syntax (the magma's non-associativity) is neurally enforced even while order-insensitive commutativity is preserved.

4. Conclusion

Happy is he who can with a vigorous wing
 Propel toward the luminous and serene realms;
 He whose thoughts, like larks,
 Free, in the morning take flight,
 — Hover over life, and understand with ease
 The language of flowers and silent things!
 Charles Baudelaire, 'Élévation,' *Les Fleurs du mal* (1861)

I have presented some avenues to build a hybrid neurosymbolic model of syntax in the brain, anchored around the ROSE neurocomputational architecture. ROSE has been shown here to be suitably flexible to accommodate different contextually driven parsing strategies across disparate cortical networks, but also mechanistically robust enough to provide a tentative answer to the question of how the brain integrates contributions from Bayesian/predictive, vector-symbolic, and oscillatory phase codes. The brain may at some fundamental level be designed for Bayes-optimal cue integration (Jordan et al., 2024), but neurolinguists have to appropriately theorize for levels of information integration that reach into higher cognitive forms of generativity.

By drawing principled relations across computational, algorithmic and implementational Marrian levels, ROSE is well positioned to provide a scaffold for what we can think of as a 'Universal Neural Grammar' (UNG) for human language – a species-general format for neurally organizing the construction of hierarchical, compositional syntactic structures (Figure 6), which matures in accordance with a genetically determined biological matrix (Nefdt, 2023; Trettenbrein et al., 2025; Yang et al., 2017). Consider here also how 'spikiness' and 'animacy' seem to be major organizational principles of ventral visual cortex (Bao et al., 2020; Coggan & Tong, 2023). What might we expect here for language's representational axes (perhaps in pMTG) that could potentially be innately encoded? Instead of spiky-stubby and animate-inanimate, we may instead have concrete-abstract, predicate-non-predicate, or conceptual-formal. This remains to be tested, but if the R component is accurately formulated it could readily accommodate these forms of representational axes.

The central thesis of this article is directly in line with proposals made in the seminal work of Ding et al. (2016),

that many linguistic 'operations may correspond to neural computations on distinct frequency scales, which are coordinated by the low-frequency neural tracking of linguistic constituent structures.' Specifically, ROSE provides a mechanistic basis for distinct neural processing timescales (Figure 6). Low-frequency synchronization is how large-scale integrative networks interface with local compositional hotspots in the language cortex (Figures 3, 4). ROSE offers one possible framework for integrating symbolic and statistical inferences, but it is relatively lean in its assumptions, and remains in an early stage of specifying how exactly statistical cues fuse with the hierarchical symbolic pipeline (though I have sketched out steps to explore this). I have focused here on mechanisms in physical space (traveling waves, population spiking), but models that incorporate state space (Glaser et al., 2020) and metastable network dynamics (Forseth et al., 2021) will likely also be needed to fully articulate the interactions between statistical inferences and symbolic state transitions. Other open questions concern the potential relevance of mechanisms that are not a core part of ROSE, like phase precession, which has been argued to bear a causal role in various cortical processing regimes (Wischniewski et al., 2024). Future work should also seek additional quantitative complexity indices for an entire ROSE parse or discourse fragment, complementary to mutual information and surprisal; e.g., ROSE might leverage the categorical/topological complexity that some authors have argued is required for rich neural computation (Manin & Marcolli, 2024).

Other recent directions in cognitive neuroscience seem to be converging on ideas sympathetic to the present proposal (Overton et al., 2025). In the neurobiology of risk/reward expectations, high-frequency γ activity is now being recognized as insufficient for coding risk preference. While γ successfully encodes choice-related information, some researchers are acknowledging that more stable, circuit-wide activity (via low-frequency oscillatory coherence) may constitute a more plausible mechanism for the representation of stable preferences. For related directions from the cognitive neuroscience of attention, see Banaie Boroujeni and Womelsdorf (2023).

How adaptable is the architecture provided by ROSE? It is entirely reasonable to expect that in the future we may be able to dispense with the notion of a low-frequency phase code altogether, and with further direct cortical recordings expose how a combination of specialized and generalized dynamical motifs can be coordinated through some other mechanism of cortical computation that can provide adequate explanatory power in more parsimonious ways. It is also possible that we may be able to dispense even with the notion

of ‘syntactic category’ (N, V, AP, etc.) and only invoke semantic types for phrase composition (Elbourne, 2025), reducing the contributions from S/E levels in driving syntax. Indeed, Elbourne’s (2025) framework may be more directly relatable to neurocognitive architectures like ROSE in which conceptual representations are primary (i.e., the system ‘starts’ with lexico-semantic features at R).

When discussing symbolic models versus neural nets, Dehaene et al. (2022) note that ‘[a] middle ground must be found.’ Although I have discussed how specific parsing strategies can in principle be implemented via ROSE, much remains unexplored. A reasonable complaint from Piantadosi (2021) is that while ‘[t]here are many connectionist approaches that try to explain – or explain away – symbolic thought,’ in contrast ‘almost no work on the symbolic side has sought to push down towards representations that could more directly be implemented’ (see Piantadosi, *forthcoming*). The hybrid model I have developed here constitutes such an example of a symbolic theory aiming to make room for connectionist processes. Hybrid models of syntax in the brain can help unify perspectives that rightly emphasize the existence of linguistic prediction (Forseth et al., 2020) with frameworks that highlight the role of language in inference generation, cognitive model updating, the construction of language-specific types of thought, consolidation of experience, endogenous monitoring, aiding directed attention, and the reflection of personal experience and values.

Developing hybrid models will enhance our ability to more confidently isolate neural mechanisms of syntax. The pluralistic research program I have proposed here is, I hope, in the spirit of Alan Turing’s intuition that codes are to be cracked not through brute force, but through the establishment of higher-order abstractions taking scope over existing models.

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